Data Science-based Techniques for Modelling and Diagnostics of Battery Cells Based on End-of-Life criteria

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2023 International Conference on Electrical Drives and Power Electronics (EDPE) | 979-8-3503-2275-0/23/$31.00 ©2023 IEEE | DOI: 10.1109/EDPE58625.2023.10274007

***Abstract*— This research work focuses on implementing Data Science techniques for Battery Energy Storage Systems (BESSs) according to Health and Charge indicators based on End-of-Life (EOL) criteria. A simple Equivalent Circuit Model (ECM) is implemented to illustrate the behavior of a battery cell in parallel with numerical methods. Density-based spatial clustering of Applications with Noise (DBSCAN), Ordering Points to identify the Clustering Structure (OPTICS), and Local Outlier Factor (LOF) are the Unsupervised techniques implemented for anomaly detection, while Multi-Layer Perceptron (MLP), Long-Short Term Memory (LSTM), and Gated Recurrent Unit (GRU) are the Supervised algorithms executed for diagnostics. Similar results are obtained in the outlier analysis, additionally, Supervised algorithms show a high level of performance and provide a basis for determining the capacity of the battery based on the End-of-Life criteria (EOL) of a Battery Energy Storage System (BESS).**

Keywords—Battery Management Systems, Machine Learning, Computer Science.

I. Introduction

The energy transition is a promising topic to achieve climate change mitigation and reduce CO2 emissions across the globe, being the point of discussion in the scientific and technological fields. Electrification composes a beneficial source of renewable energy to substitute fossil fuels with electricity, not only decreasing the quantity of air pollutants but also offering more affordable prices that satisfy consumer needs. In the operation of an electric vehicle, a Battery Energy Storage System (BESS) is a core element in the mechanism of working to provide reliable and safe transportation, however, specific methodologies must be considered to achieve optimal performance.

The build, design, and operation are crucial steps to monitor the performance of a BESS, which are based on the Key Performance Indicators (KPIS) during the modelling implementation. Gilbert Zequera et al. [1] conducted a literature review about the different types of battery models for Electric Vehicles (EV) and Digital Twin (DTs) applications, being Equivalent Circuit Model (ECM), Mathematical Model, and Electrochemical Model, all the basis knowledge for developing, verifying, and implementing a battery diagnostic.

Health and Charge indicators play an important role not only in battery modelling, but also in fault diagnostics and predictive maintenance [2], so that conducting experimental measurements and testing processes are additional tools in the life cycle of a BESS. The State of Health (SOH) is defined as an indicator that measures the life cycle of a battery in comparison with its initial capacity as well as its degradation level, which has encouraged the development of new Data-Driven and statistical approaches [2,3].

In the computer science field, Data Science techniques provide fast and reliable solutions to manage batteries and their applications in the renewable energy industry, however, it is important to point out that specific algorithms must be considered to satisfy user needs based on several restrictions. The End-of-Life (EOL) criteria is defined as a concept in a BESS that explains the failure in the performance or functionality, which is usually associated with 70-80% of the total rated capacity [4].

In this article, a dataset based on lithium-ion battery cycling tests is presented to evaluate robustness of various Data-Driven methods. The main goal of current research work is to provide modelling and diagnostics of battery cells through the implementation of Data Science-based techniques, all to initialize a research environment in battery management for DTs and EVs applications.

The rest of the paper is organized as follows, in Section II, the problem statement, dataset, and the motivation of this research are explained through the implementation of a battery model. Section III describes and executes Unsupervised techniques, focusing on outlier analysis. Section IV implements Supervised techniques to estimate the battery capacity based on the EOL criteria. Section V discusses the results, and finally, in Section VI a conclusion is presented to encourage the continuation of this work by considering more advanced battery models and computer science algorithms for diagnostics of different cells and packs.

II. Battery Modelling and Problem Statement

The ECM is defined as a Grey-box model that provides understanding and ease access to parameters in a BESS,

The research has been supported by the Estonian Research Council under grant PSG453 "Digital twin for propulsion drive of an autonomous electric vehicle".

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providing a medium time-consuming modelling that reflects the behavior under an engineering framework [1].

Mathematically, the ECM is integrated by several parameters, such as the State of Charge (SOC), Voltage source (V_{oc}), Voltage across the Solid Electrolyte Interface (SEI), and cell voltage output $v_b(t)$. The SOC and voltage source are represented by the following equation [5]:

$$SOC(t) = SOC_0 + \frac{1}{Q}\int_0^t i_b(t) - \frac{V_{oc}(SOC(t))}{R_d}dt\,, \qquad (1)$$

The variable $Q$is defined as the capacity of the battery, R_d is the self-discharge resistor, SOC_0 is the initial State of Charge, and $i_b$is the applied current. Similarly, the voltage across the SEI is represented as follows [5]:

$$\Delta v_t(t) = \frac{1}{C_t}\int_0^t i_b(t) - \frac{\Delta v_t(t)}{R_t}dt\,, \qquad (2)$$

The terms $C_t$refers to the double layer capacitance, and $R_t$is the resistance of the SEI layer charge transfer Finally, the cell output is represented by the total sum of the voltages [5]:

$$v_b(t) = V_{oc}\big(SOC(t)\big) + i_b(t)R_s + \Delta v_t(t)\,, \qquad (3)$$

A simple ECM has been implemented in Python, in which the input parameters specified in the previous equations have been defined to compute the output variables. A graphical representation is illustrated in Fig. 1, considering the SOC and Voltage source.

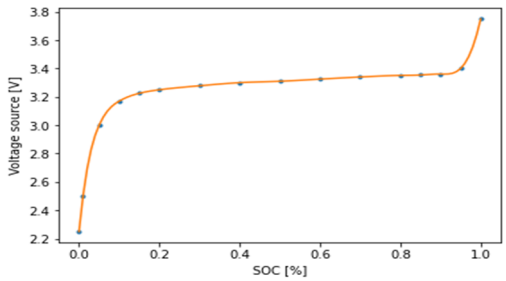

Fig. 1. ECM representation of the Voltage source vs SOC.

Additionally, numerical approximations can be executed to achieve a high level of accuracy in a battery model, by comparing the predicted results with the measured voltage from a BESS. Solution of linear and non-linear equations can be solved through dynamic optimization to fit experimental data in regression problems. The GEKKO Optimization Suite [6] has been implemented to solve and fit the equations of the ECM, which illustrates the results of the cell voltage output in Fig 2.

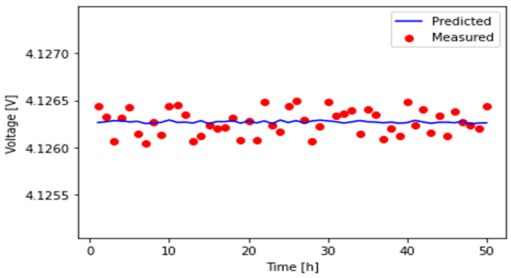

Fig. 2. Numerical approximations to fit experimental data in the ECM, using the GEKKO Optimization Suite.

It is important to mention that battery modelling explains the physics of the system based on mathematical equations so that the explanation and calculation of the KPIs are the next steps to provide the diagnostics of a BESS. The problem statement in the next sections corresponds to implementing Unsupervised and Supervised techniques to achieve diagnostics of a batter cell, all by considering the trends and crucial values of the Health and Charge indicators based on the EOL criteria.

The dataset used in this article corresponds to an experimental test of two prismatic lithium-ion battery cells containing SOH as the predicted variable and the features used for estimations, which are the Voltage, Current, Resistance, Time, Charge capacity and Discharge capacity. Cells were tested in a charge profile that was a standard constant current/constant voltage protocol with a constant current rate of 0.5C until the voltage reached 4.2V, and then held 4.2V until the charging current dropped below 0.05A. The discharge cutoff voltage for these batteries was 2.7V and the total capacity of the cells is 1.1 Ah. Additional information about the dataset can be found in reference [7].

Motivation of this research is based on the development of an initial methodology that can provide a research environment for modelling and diagnostics of a BESS, all by implementing Data-Science based solutions and combining the knowledge from both fields, engineering, and Machine Learning (ML).

III. Unsupervised Techniques

Initially, the dataset is processed, and Feature Engineering is executed to calculate the new features, which are the Constant Current Constant Time (CCCT), Constant Voltage Constant Time, Cycles, Discharge capacity, and the State of Health (SOH), the latter being defined as the predicted variable. The next step consists of implementing Exploratory Data Analysis (EDA) to understand the correlation of every feature and the corresponding KPIs. Subsequently, some parameter estimation techniques are initialized to start the Unsupervised techniques and achieve outlier analysis. Fig. 3 illustrates the KPIs after executing the Feature Engineering process in the dataset, representing the CCCT, CVCT, Internal resistance, Capacity, and the SOH for the battery cells.

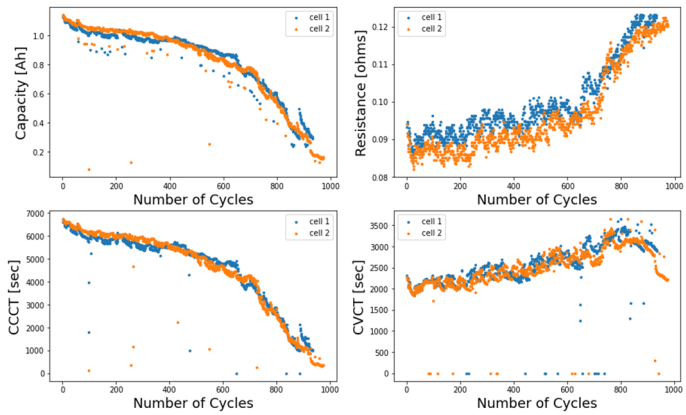

Fig. 3. Initial plots after implementing Feature Engineering in the battery cells. The upper plot represents the Capacity and Resistance, while the lower plots indicate the CCCT and the CVCT.

A. Parameter optimization and initialization

Density-based spatial clustering of Applications with Noise (DBSCAN) is defined as a density-based clustering algorithm whose input parameters of performance are Eps and MinPts. Eps denotes the maximum distance between two points, while MinPts is the minimum quantity of selected

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points in the neighborhood. Determination of the optimal Eps value is based on a single-level density approach that calculates the slope of 1% difference between the points of the k-nearest neighbor [8]. Regarding the optimal MinPts in the DBSCAN implementation, a heuristic approach that consists of the k-th nearest neighbor distance function in parallel with a density distribution in the dataset is considered [9]. The methodology to determine the optimal Eps value is exemplified in Fig. 4.

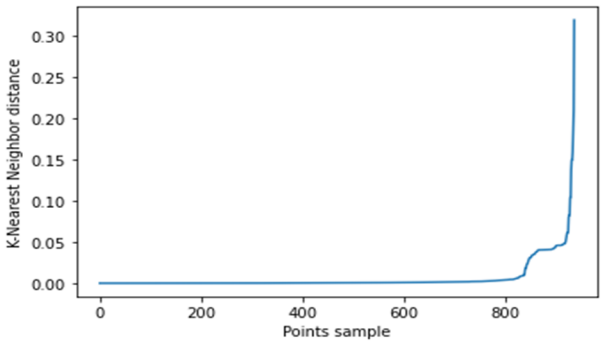

Fig. 4. Determination of the optimal Eps based on the k-Nearest Neighbor approach through a single-level density algorithm.

Local Outlier Factor (LOF) is defined as a normalized density-based approach used for anomaly detection. LOF can detect outliers by comparing the density of each data point with the density of its respective k-nearest neighbor, thus input parameters are integrated by the minimum number of points "q" and the threshold "p". Selection of the optimal "q" value can be determined by choosing a maximum and minimum number of points, subsequently, taking the maximum value over each "q" in a specified interval until convergence is achieved [10]. Threshold value is selected by considering the LOF score distribution and calculating the quantile point, finally, the threshold is adjusted based on the user criteria. In Fig. 5, the optimal threshold has been determined by using several LOF models based on the distribution of the LOF score over a total range of 200 Nearest-Neighbors.

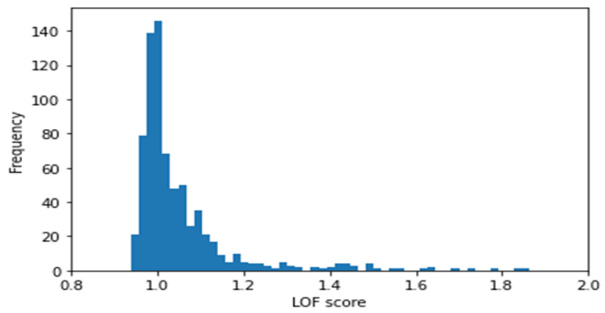

Fig. 5. Threshold determination of the LOF based on aggregating multiple LOF models and score distribution. 20 LOF models are selected

Ordering Points to identify the Clustering Structure (OPTICS) is defined as a density-based clustering method that has the potential to address data points of different densities, being considered as an extension of DBSCAN. Implementation of the OPTICS algorithm introduces an additional distance for each data point, which not only represents the density but also is used as selected criteria to determine if a point is considered a cluster. The input parameters have the same meaning as DBSCAN, which are the cluster radius and a minimum number of points; however, OPTICS has the advantage of overcoming density limitations, primarily based on using core distance, reach distance, and spawn distance [11].

B. *Outlier analysis*

Implementation of the Unsupervised techniques has been achieved by using the appropriate parameter optimization techniques explained in the previous subsection. Regarding the detection of outliers, results indicate that the threshold plays the most important role in determining if a point is considered or not as an outlier, which not only increases or decreases the quantity of outliers in the dataset but also specifies the interval value for each feature.

Due to the mutual correlation between the capacity of the battery and the SOH, the focus is based on identifying the intervals that have a major quantity of outliers during the EOL criteria, all to provide assessment during the data processing and avoid possible potential biases during the battery diagnostics. Outlier analysis of the LOF, DBSCAN, and OPTICS for the first cell is shown in Fig. 6

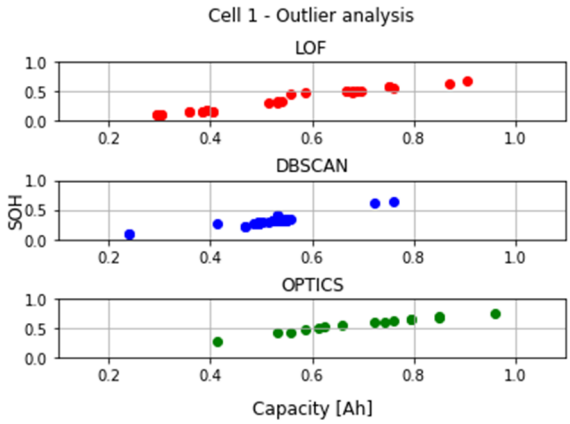

Fig. 6. Outlier analysis representation by the LOF, DBSCAN, and OPTICS in the SOH vs Capacity curves for the first cell.

It can be appreciated that for the first cell, outliers are generally close to the EOL criteria, demonstrating that the implemented algorithms provide robust performance, additionally, LOF and OPTICS show similar behavior during outlier detection, specifically due to the outlier range when SOH is around 60-70%.

Results of the second cell are shown in Fig. 7, illustrating a similar pattern in the SOH vs Capacity curve, however, compared to the first cell, a higher quantity of outliers is found, which indicates a possible anomaly behavior at some points during the battery operation.

According to the results illustrated in Fig. 7, LOF and OPTICS not only show a higher quantity of outliers than DBSCAN, but also detect different intervals in the SOH vs Capacity curves. On the other hand, DBSCAN results indicate a remarkable distribution of outliers into specific groups, corresponding to intervals below the EOL criteria, which differs from the outlier distribution of both OPTICS and LOF.

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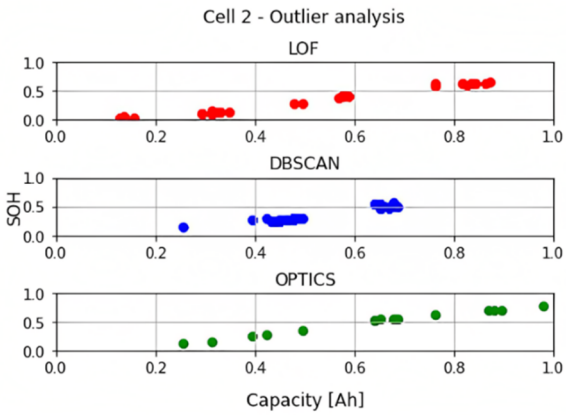

Fig. 7. Outlier analysis representation by the LOF, DBSCAN, and OPTIICS in the SOH vs Capacity curves for the second cell.

Before concluding this section, it is important to point out that for the implementation of Supervised techniques, those points detected as outliers have been removed with the purpose of achieving a remarkable accuracy in the performance metrics. A more detailed analysis will be provided in the results subsection at the end of this article.

IV. Supervised Techniques

In this section, Supervised techniques are implemented in the dataset after performing Feature Engineering, EDA, and outlier analysis. The main goal of this section is to implement, evaluate, and compare the performance of different ML methods, all to estimate the capacity of the battery based on the calculated KPIs and EOL criteria.

Multi-layer Perceptron (MLP) is defined as an Artificial Neural Network (ANN) process, also considered as a complement of a feed-forward neural network that consists of three layers: an input layer, a hidden layer, and an output layer. Each neuron in the hidden layer is connected to the neurons in the next layer. The connecting wires between the neurons are known as weights whose values are updated with the help of the learning phase. The learning phase is continuously repeated until the error value is less than the threshold level. The input layer is the combination of the values of the features and the output layer is responsible for functions like classification and prediction. It is necessary to mention that the directly connected mechanism of MLP consists of an infinite series of hidden layers located between the output and input layers. Finally, in a MLP the data passes in a forward path from the input to the output layer, being the equivalent of a feed-forward that uses backpropagation to train all the nodes [12,13].

Long Short-Term Memory (LSTM) is a Recurrent Neural Network (RNN) whose operating mechanism consists of a more sophisticated process, which is composed of three gates (forget, input, and output) that regulate the information flow in the neural network unit for long periods of time. Compared to an RNN, an LSTM not only provides more efficient performance in complex tasks such as language modelling and language translation but also resolves the vanishing gradient problem that an RNN is not able to solve. The vanishing gradient problem occurs when a zero gradient is obtained, and the weights cannot be optimized during the backpropagation process, therefore, the amount of data that can be stored in the memory is limited. LSTM-type RNN solves the vanishing gradient problem by implementing the three gates to decide how much of the memory should be allocated. Additionally, the problem is solved by regulating not only the quantity of data that gets into each corresponding time step but also the quantity of weights that are optimized [14, 15].

Gated Recurrent Unit (GRU) is a type of RNN that was introduced in 2014 by Cho et al. as a simpler alternative algorithm to LSTM. GRU has not only the potential to make each recurrent unit adaptively capture dependencies of different time scales but also an operating mechanism that uses less training parameters and memory. Like the LSTM unit, the GRU has gating units that modulate the flow of information inside the unit, however, without having a separate memory cell. Among the remarkable applications of the GRU are the well performance and accuracy in sequence learning tasks, overcoming the problems of vanishing and explosion of gradients in traditional recurrent neural networks (RNNs) when learning long-term dependencies. The architecture of a GRU is based on an input layer composed of multiple neurons, the number of neurons is determined by the size of the feature space. It is necessary to point out that the number of neurons in the output layer corresponds to the output space. Hidden layers containing memory cells cover the main functions of GRU networks. Cell state changes and maintenance depend on two gates in the cell: a reset gate and an update gate [16, 17, 18].

A. *Cell diagnostics*

MLP, LST, and GRU are the selected and implemented algorithms to estimate the capacity of the battery. As mentioned at the end Section III, outliers have been removed, and the dataset is processed to initialize the Supervised techniques. It is important to point out that the main purpose of this approach is to implement and compare the performance of different ML algorithms that optimize the operation of the battery cells so that estimations of the SOC and SOH can be automatically performed.

In this specific case, the rated capacity of each cell has the value of 1.1 Ah, therefore the EOL at 70% of the battery cells is reached at approximately 0.77 Ah. A new visualization of the battery capacity vs number of cycles for two cells is shown in Fig. 8, which differs from Fig. 3 due to the execution of outlier analysis.

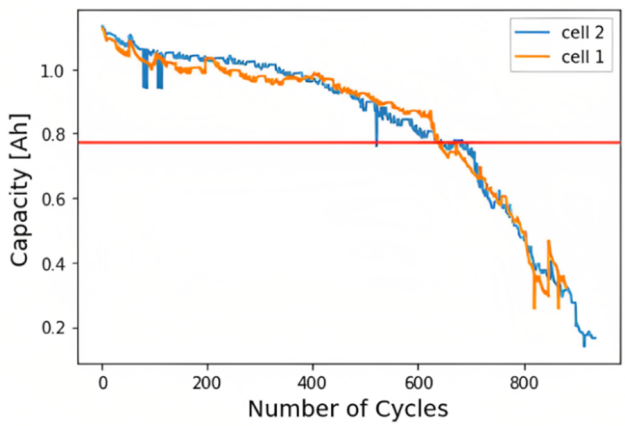

Fig. 8. Remaning capacity vs number of cycles for two battery cells. Horizsontal red line denotes the EOL criteria specied at 70% of the total rated capacity.

Diagnosis of the battery cells is based on a Data Science approach and is summarized in the following points:

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- The dataset is pre-processed and EDA in parallel with Feature Engineering are executed. During the Feature Engineering process, each variable is separated into charging and discharging. Additionally, the CCCT, CVCT, discharge capacity, and SOH are the new calculated KPIs based on the initial features (e.g., current, voltage, and time). The dataset of each battery cell is updated according to the status profile and each measurement.
- Unsupervised techniques are executed. LOF, DBSCAN, and OPTICS are implemented to achieve outlier analysis, first, parameter optimization is initialized, second, each unsupervised technique with different hyperparameters is validated to test several models, finally, the outliers are detected and dropped from the dataset.
- Helper functions are coded to achieve data processing. The dataset is divided into training, validation and testing, performance metrics are executed to verify the accuracy of the validation process, finally, the Neural Networks modules are compiled and customized based on the battery cells to initialize the training process.
- Hyperparameters are selected to initialize the training and validation processes. Learning rate, epochs, dimension of hidden layers, and window size, are the most crucial hyperparameters for the Supervised techniques. Training and validation loss for each epoch are provided to verify the suitability of the corresponding models, preventing biases, underfitting, and overfitting.
- Supervised techniques are implemented to conclude the Data Science approach. After executing the training and validation processes, the testing set is used to make predictions of the battery capacity. Performance metrics for each model are provided to compare the accuracy, finally, the estimations and true values are visualized.

Graphical results of the diagnostics are provided in Fig. 9, comparing the predictions with the true values of the dataset.

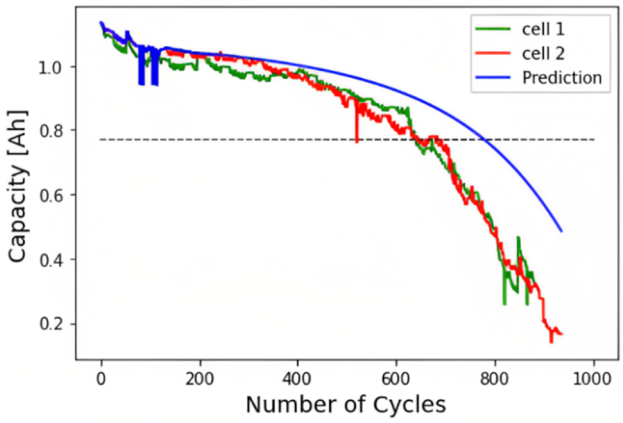

Fig. 9. Graphical results of the Supervised techniques in the Capacity vs Number of cycles for two cells . Horizsontal line denotes the EOL criteria specied at 70% of the total rated capacity.

V. Performance Metrics and Discussion

The performance of the Supervised techniques for both battery cells is evaluated in the testing process. Numerical results of the Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Relative Error (RE) are shown in Table 1.

TABLE I. PERFORMANCE METRICS

Supervised technique	*RMSE*	*MAE*	*RE*
MLP	0.1140	0.0908	0.1430
LSTM	0.0993	0.0785	0.1395
GRU	0.0822	0.0614	0.0989

All the Supervised techniques show a similar performance in the Testing step, corroborating the Training and Validation loss over the total of epochs, however, compared to MLP, the RNNs provide a more optimal performance, which in this specific problem not only prevents overfitting, but also a possible bias due to model simplicity or erroneous assumptions.

Hyperparameter tuning and Validation play the most important role in implementing the diagnostics and achieving accurate estimations of the predicted variable, all because of the calculated KPIs in the data preprocessing. Furthermore, Feature Engineering is a helpful process that can provide promising insights not only to analyze the data set, but also to obtain a new subset of features in the operation of a BESS.

Regarding the Unsupervised techniques, outlier analysis is a useful tool that not only improves the accuracy of the Data Science approach on performance metrics, but also helps to understand a possible anomaly in the dataset during battery operation. Compared to DBSCAN, LOF and OPTICS provide similar results in outlier detection, which rely on the properties of each algorithm to identify data points with variable density and hidden irregular patterns in various datasets. The LOF implementation has less complexity than density-based methods, making this method easy for the user to understand; however, the threshold criterion is a key factor in categorizing a data point as a potential outlier. To obtain more reliable results in battery datasets with different KPIs and high feature dimensionality, OPTICS overcomes the limitations of DBSCAN by using the concept of reachability distance, which adapts to the local density of the data.

According to the properties of density-based methods, outlier analysis is one of the possible drawbacks due to the possibility of dealing with variable densities. Additional drawbacks of unsupervised techniques are the difficulties in evaluating results due to the lack of a universal measure of success or failure. However, one way to evaluate the technique is to compare the results with the domain knowledge or using error analysis and performance auditing, which rely on quantitative metrics, human level performance, visual or qualitative methods.

Based on the properties of battery datasets, their initial features, and the calculated KPIs, RNNs are beneficial for handling sequential or temporal information in cell diagnostics, so that data points are used in sequence to make

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accurate predictions. The RNN implementation introduces an internal loop in the number of cycles that allows the battery predictors to be processed from one step of the network to the next one. Examples of the suitable tasks for the RNNs under the framework of a BESS are: 1) variable length of sequences in the charge and discharge, 2) share parameters across the sequences of battery cycles, 3) maintain a sequence of order in the KPIs and 4) keep track of long-term dependencies in the battery capacity. MLP showed a lower level of performance than the RNNs methods, supporting the previous explanation about the implementation of sequential modelling in cell diagnostics. Due to a lower level of complexity in the modelling and architecture mechanism, GRU is a more recommended algorithm compared to LSTM for this dataset, however, a small difference in the metrics of performance was obtained, concluding that similar results can be obtained through RNNs. Finally, drawbacks in the implementation of RNNs are the computational and training complexity, prone to overfitting with noisy or small datasets, and possibility to experience either vanishing gradient or exploding gradient problems.

VI. Conclusion

Data science-based techniques are described, discussed, and implemented to provide diagnostics of battery cells based on EOL criteria. The objective of the research work was to familiarize the reader with an assessment of battery cells by explaining the physics behavior through an ECM and then providing diagnostics using ML algorithms. Regarding the implementation of both Supervised and Unsupervised techniques, parameter optimization and validation must be considered to achieve optimal metrics of performance in the Testing step. Therefore, it is crucial to become familiar with the mathematical foundations of the computational algorithm to understand the behavior of the battery in the different stages of the battery life.

The novelty associated with the proposed methods is based on the implementation of robust computer science algorithms that not only perform Data Science techniques to estimate Health and Charge indicators but also consider battery modelling based on the physics framework to provide a meaningful insight about the build, design, and operation of a BESS. Innovative techniques are summarized to obtain helpful insights into the KPIs in battery cells, all to demonstrate that developing, verifying, and implementing physics models of battery systems are beneficial to achieve the energy transition.

Different types of battery cells must be tested in experimental measurements, thus, it is necessary to compare the behavior of battery models than can explain the dynamics of the entire system, such as ECM, Mathematical Model, and Electrochemical Model are further points of discussion to encourage future research. This work is a basic methodology for the diagnostics of BESS through ML network architectures by the utilization of Explainable Artificial Intelligence methods. Moreover, this work provides an initial research environment for battery management in DTs and EVs applications to test and compare the performance of battery cells. This will help establish assessment and verification procedures for possible fault diagnostics to support commercial consulting, research, and testing for enterprises based on the DT concept.

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